



2. Strong interaction



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1. Deep inelastic scattering

the γ interacts with one constituent of the proton

→ historically called partons (Feynman)



for high-energy scattering

$$E \gg m_p$$

the struck parton carries a fraction of the total momentum

of the proton

$$x = \frac{Q^2}{2p \cdot q} \quad 0 \leq x \leq 1$$

$\nwarrow$ momentum of the γ^*
 $\nearrow$ momentum of the proton

parton distribution function (PDF)

for the up-quark PDF

$$u(x) \delta(x)$$

↑
determined experimentally

⇒ number of up-quarks that have a momentum fraction between

$$x \text{ and } x + \delta x$$

PDF cannot be computed from theoretical calculation

⇒ non-perturbative regime

⇒ PDF are determined experimentally

electron-target exp
 pp collider LHC
 pp̄ collider Tevatron

very high energy colliders

$$\sqrt{s} \gg 1$$

the fraction of partons with low x increases

Illustration